Fall 2025 CPTS530 Midterm Project
MATH 448-548
Numerical Analysis

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Contents

Problem 1

Problem Statement: Use Taylor theorem to derive the error term for the approximate formula

$$f'(x) \approx \frac{1}{2h}(-3f(x) + 4f(x+h) - f(x+2h)).$$

Solution:
We are given the finite-difference formula:

$$f'(x) \approx \frac{1}{2h}\left[-3f(x) + 4f(x+h) - f(x+2h)\right]$$

Expand $f(x+h)$ and $f(x+2h)$ about x using Taylor's theorem, keeping terms up to h^2 and representing higher-order terms as k_1h^3 and k_2h^4:

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + k_1h^3$$
$$f(x+2h) = f(x) + 2hf'(x) + 2h^2f''(x) + k_2h^3$$

where k_1, k_2 are finite constants depending on third order derivatives of f at x.

Substitute into the formula:

$$\begin{aligned}
&\frac{1}{2h}\left[-3f(x) + 4f(x+h) - f(x+2h)\right] \\
&= \frac{1}{2h}\Bigg[-3f(x) + 4\left(f(x) + hf'(x) + \frac{h^2}{2}f''(x) + k_1h^3\right) \\
&\qquad - \left(f(x) + 2hf'(x) + 2h^2f''(x) + k_2h^3\right)\Bigg]
\end{aligned}$$

Simplifying the terms in the square brackets:

$$= \frac{1}{2h}\left[2hf'(x) + (4k_1 - k_2)h^3\right]$$

Now, divide each term by $2h$:

$$= f'(x) + \frac{1}{2}(4k_1 - k_2)h^2$$

Thus, the leading error term is $O(h^2)$:

$$\text{error} \sim O(h^2)$$

This shows the method is second-order accurate. Note that I've already checked if the third order term vanishes if we write out the taylor's polynomial to the third order, and it does not.

Problem 2

Problem Statement: Consider the following variation of the Newton's method:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_0)}.$$

Find constants C and s such that

$$e_{n+1} = Ce_n^s.$$

Solution:

Problem 3

Problem Statement: Consider the linear system $\mathbf{Ax} = \mathbf{b}$, where

$$\mathbf{A} = \begin{bmatrix} 6.25 & -1 & 0.5 \\ -1 & 5 & 2.12 \\ 0.5 & 2.12 & 3.6 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 7.5 \\ -8.68 \\ -0.24 \end{bmatrix}.$$

Write a computer program for LU-factorization with a unit lower triangular $\mathbf{L}$ (meaning that the diagonal entries should be equal to one). Then write a program for the Cholesky factorization.

WARNING: avoid using shortcuts. The programming should be done "from scratch".

Solution:

Problem 4

Problem Statement: Consider the equation

$$f(x) = 0,$$

where

$$f(x) = x - \frac{1}{2}\left(\cos(x/2) - \left|x - \frac{1}{2}\right|\right).$$

Do the following.

Part 1

Rewrite the equation as a fixed point problem for a function $T(x)$ related to $f(x)$. Prove that $T(x)$ (i) maps the interval $[0.45, 0.55]$ into itself, and (ii) $T(x)$ is contractive on this interval. Based on the theorems in the books and notes, make conclusions about existence and uniqueness of a solution to $f(x) = 0$.

Solution:

Problem Statement: Consider the equation

$$f(x) = 0,$$

where

$$f(x) = x - \frac{1}{2}\left(\cos(x/2) - \left|x - \frac{1}{2}\right|\right).$$

Do the following.

Part 2

Write the computer program implementing the fixed point iteration $x_{n+1} = T(x_n)$. Discuss the choice of an initial guess x_0. Calculate the solution to within 10 decimal places. Note the number of iterations needed.

Solution:

Problem Statement: Consider the equation

$$f(x) = 0,$$

where

$$f(x) = x - \frac{1}{2}\left(\cos(x/2) - \left|x - \frac{1}{2}\right|\right).$$

Do the following.

Part 3

Use a priori and a posteriori error estimates from the notes to estimate the number of iterations needed to obtain the above accuracy. Which estimate is more accurate?

Solution:

Problem Statement: Consider the equation

$$f(x) = 0,$$

where

$$f(x) = x - \frac{1}{2}\left(\cos(x/2) - \left|x - \frac{1}{2}\right|\right).$$

Do the following.

Part 4

Implement the Newton's method on a computer for the same equation starting with x_0 within the interval $[0.45, 0.55]$. Discuss the theoretical difficulty related to non-existence of $f'(x)$ for certain point(s) within the interval. How would you set up the Newton's method to resolve the problem?

Solution:

Problem Statement: Consider the equation

$$f(x) = 0,$$

where

$$f(x) = x - \frac{1}{2}\left(\cos(x/2) - \left|x - \frac{1}{2}\right|\right).$$

Do the following.

Part 5

Note the number of iterations needed to calculate the solution to within 10 decimal places using the Newton's method. Discuss the speed of convergence of the Newton's method and compare it with the convergence speed of the fixed point iterations.

Solution:

Problem Statement: Consider the equation

$$f(x) = 0,$$

where

$$f(x) = x - \frac{1}{2}\left(\cos(x/2) - \left|x - \frac{1}{2}\right|\right).$$

Do the following.

Part 6

Implement the alternative error control strategy proposed by you in HW 3, Problem 2. Compare the results with the ones obtained using a priori and a posteriori error estimates.

Solution:

Problem 5

Problem Statement: Implement and analyze the Ridders' method for solving the equation $f(x) = 0$. For guidance, you might wish to read the Wikipedia page on the Ridders' method. Also feel free to consult other sources.

Brief Description of the method

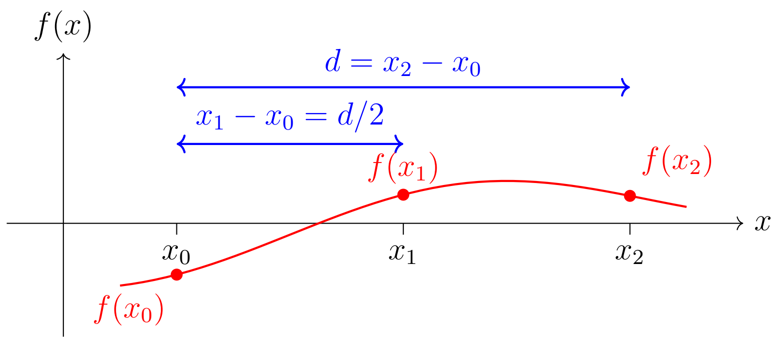
Given the initial bracketing interval $[x_0, x_2]$, containing **one** solution, and such that $f(x_0)$ and $f(x_2)$ have opposite signs, compute the midpoint $x_1 = (x_0 + x_2)/2$. Then define a new function $h(x) = f(x)e^{ax}$. Find the parameter a that ensures $h(x_1) = \frac{1}{2}(h(x_0) + h(x_2))$. Then define x_3 as the x-intercept of the line passing through $(x_0, h(x_0))$ and $(x_2, h(x_2))$. Use x_3 as one of the endpoints of the next interval bracketing the solution. The other endpoint is x_1 if $f(x_1)f(x_3) < 0$. Otherwise, choose either x_0 or x_2 based on the requirement that the sign of $f(x)$ at the chosen point must be opposite to the sign of $f(x_3)$. Continue until the desired accuracy is reached.

Part 1

Show that a is uniquely defined and give the equation for finding it.

Solution:

Note: Thanks to the wikipedia article which itself cites the original paper by Ridders, it was easier to derive all the relevant formulae. I have chosen my target formulae and required derivations to align with those, with little variation.



Let x_0 and x_2 be the endpoints of the bracketing interval, and $x_1 = \frac{x_0+x_2}{2}$ the midpoint. Let $d = x_2 - x_0$.

Define $f_0 = f(x_0)$, $f_1 = f(x_1)$, $f_2 = f(x_2)$.

Ridders' method introduces a function $h(x) = f(x)e^{ax}$, and chooses a so that:

$$h(x_1) = \frac{1}{2}(h(x_0) + h(x_2))$$

$$f_1 e^{ax_1} = \frac{1}{2}\left(f_0 e^{ax_0} + f_2 e^{ax_2}\right)$$

We know that $x_1 = x_0 + \frac{d}{2}$ and $x_2 = x_0 + d$.

Substitute and rearrange:

$$f_1 e^{a(x_0+d/2)} = \frac{1}{2}\left(f_0 e^{ax_0} + f_2 e^{a(x_0+d)}\right)$$

Dividing both sides by e^{ax_0} :

$$2f_1 e^{ad/2} = f_0 + f_2 e^{ad}$$

$$2f_1 e^{ad/2} - f_2 e^{ad} - f_0 = 0$$

Let $y = e^{ad/2}$, so $e^{ad} = y^2$.

This gives a quadratic in y:

$$f_2 y^2 - 2f_1 y + f_0 = 0$$

Solving for y:

$$y = \frac{2f_1 \pm \sqrt{(2f_1)^2 - 4f_2 f_0}}{2f_2}$$

$$y = \frac{f_1 \pm \sqrt{f_1^2 - f_2 f_0}}{f_2}$$

Therefore,

$$a = \frac{2}{d}\ln\left(\frac{f_1 \pm \sqrt{f_1^2 - f_2 f_0}}{f_2}\right)$$

We note that since $y = e^{ad/2}$, its value must be nonnegative. Since clearly $f_1^2 - f_2 f_0 > f_1^2$ ($f_2 * f_0 < 0$), the sign is chosen so that y is positive:

$$\text{If } f_2 > 0: \quad a = \frac{2}{d}\ln\left(\frac{f_1 + \sqrt{f_1^2 - f_2 f_0}}{f_2}\right)$$

$$\text{else If } f_2 < 0: \quad a = \frac{2}{d}\ln\left(\frac{f_1 - \sqrt{f_1^2 - f_2 f_0}}{f_2}\right)$$

Or, more concisely:

$$a = \frac{2}{d}\ln\left(\frac{f_1 + \operatorname{sgn}(f_2)\sqrt{f_1^2 - f_2 f_0}}{f_2}\right)$$

or equivalently:

where $\operatorname{sgn}(f_2)$ is the sign of f_2.

Problem Statement: Implement and analyze the Ridders' method for solving the equation $f(x) = 0$. For guidance, you might wish to read the Wikipedia page on the Ridders' method. Also feel free to consult other sources.

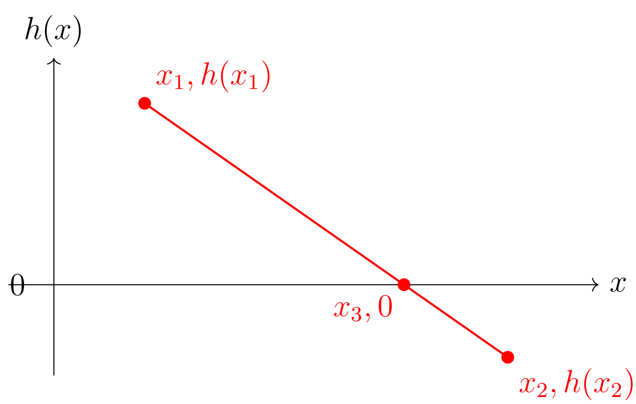
Brief Description of the method

Given the initial bracketing interval $[x_0, x_2]$, containing **one** solution, and such that $f(x_0)$ and $f(x_2)$ have opposite signs, compute the midpoint $x_1 = (x_0 + x_2)/2$. Then define a new function $h(x) = f(x)e^{ax}$. Find the parameter a that ensures $h(x_1) = \frac{1}{2}(h(x_0) + h(x_2))$. Then define x_3 as the x-intercept of the line passing through $(x_0, h(x_0))$ and $(x_2, h(x_2))$. Use x_3 as one of the endpoints of the next interval bracketing the solution. The other endpoint is x_1 if $f(x_1)f(x_3) < 0$. Otherwise, choose either x_0 or x_2 based on the requirement that the sign of $f(x)$ at the chosen point must be opposite to the sign of $f(x_3)$. Continue until the desired accuracy is reached.

Part 2

Work out a formula for x_3 in detail.

Solution:



From the diagram, we want to find x_3, the x-intercept of the line passing through $(x_1, h(x_1))$ and $(x_2, h(x_2))$.

Let $h_1 = h(x_1)$ and $h_2 = h(x_2)$. The slope of the line is:

$$m = \frac{h(x_1) - h(x_2)}{x_1 - x_2} = \frac{h_1 - h_2}{x_1 - x_2}$$

The equation of the line passing through (x_1, h_1) with slope m is:

$$h(x) - h_1 = m(x - x_1)$$

To find x_3, set $h(x_3) = 0$:

$$0 = h_1 + m(x_3 - x_1)$$

Solving for x_3:

$$m(x_3 - x_1) = -h_1$$

$$x_3 - x_1 = -\frac{h_1}{m}$$

$$x_3 = x_1 - \frac{h_1}{m}$$

Substitute the expression for m:

$$x_3 = x_1 - \frac{h_1}{\frac{h_1 - h_2}{x_1 - x_2}}$$

$$x_3 = x_1 - \frac{h_1(x_1 - x_2)}{h_1 - h_2}$$

Now, using $d/2 = x_2 - x_1$, we have:

$$\boxed{x_3 = x_1 + \frac{d}{2} \cdot \frac{1}{1 - h_2/h_1}}$$

Or equivalently:

$$\boxed{x_3 = \frac{x_1 h_2 - x_2 h_1}{h_2 - h_1}}$$

Problem Statement: Implement and analyze the Ridders' method for solving the equation $f(x) = 0$. For guidance, you might wish to read the Wikipedia page on the Ridders' method. Also feel free to consult other sources.

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Part 3

Implement Ridders' method on a computer together with the standard bisection method for finding **all** solutions of

$$e^x - x^2 = 0.$$

Solution:

First I'll plot the graph of $e^x - x^2$ to identify the intervals where the function changes sign, indicating the presence of roots.

From Figure 3, it can be seen that there is one root in the interval [-0.8, -0.6]. Since $f(x) < 0$ for $x =$ -0.8 and $f(x) > 0$ for $x =$ -0.6, I can use this interval to find the root using both the bisection method and Ridders' method.

Using the bisection method (code may be found in my solution for Hw02 Computer Problem 3.1.3), I found the root to be $r =$ -0.7034667969 to 10 decimal places, which took 12 iterations.

A screenshot of the results may be viewed in Figure 4.

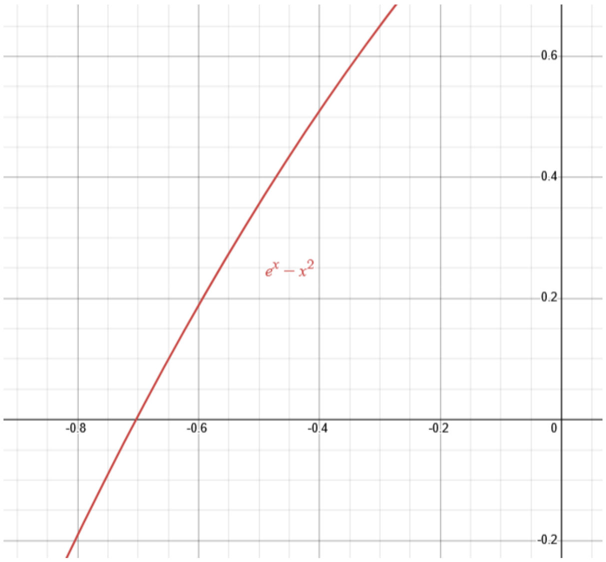

Figure 1: *
Zoomed in

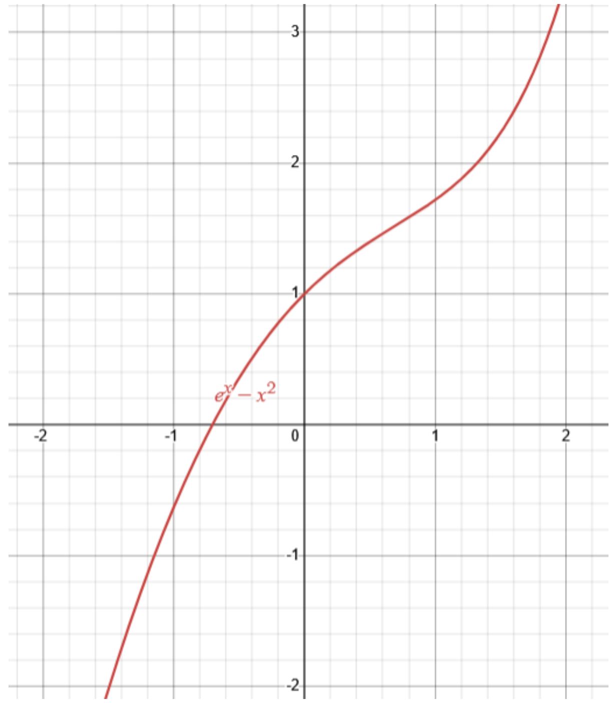

Figure 2: *
Zoomed out

Figure 3: Graphs of $e^x - x^2$: We can see from the zoomed in graph on the left that there is a single root in [-0.8, -0.6]. The zoomed out graph on the right shows the behavior of the function over a larger interval.

```
itr=1, a=-0.8, b=-0.6, c=-0.7, f(c)=0.006585303791409591, |b-a|=0.20000000000000007
itr=2, a=-0.8, b=-0.7, c=-0.75, f(c)=-0.09013344725898531, |b-a|=0.10000000000000003
itr=3, a=-0.75, b=-0.7, c=-0.725, f(c)=-0.04130043104463754, |b-a|=0.05000000000000002
itr=4, a=-0.725, b=-0.7, c=-0.7125, f(c)=-0.017239627924093892, |b-a|=0.02500000000000001
itr=5, a=-0.7125, b=-0.7, c=-0.70625, f(c)=-0.005297738100147331, |b-a|=0.012500000000000004
itr=6, a=-0.70625, b=-0.7, c=-0.703125, f(c)=0.0006511313011985931, |b-a|=0.006250000000000002
itr=7, a=-0.70625, b=-0.703125, c=-0.7046875, f(c)=-0.002321465341744766, |b-a|=0.003125000000000001
itr=8, a=-0.7046875, b=-0.703125, c=-0.7039062500000001, f(c)=-0.0008347076237052442, |b-a|=0.00156250000000000
5
itr=9, a=-0.7039062500000001, b=-0.703125, c=-0.7035156250000001, f(c)=-9.167332685461327e-5, |b-a|=0.00078125000
000000003
itr=10, a=-0.7035156250000001, b=-0.703125, c=-0.7033203125, f(c)=0.0002797576939281843, |b-a|=0.00039062500000
00013
itr=11, a=-0.7035156250000001, b=-0.7033203125, c=-0.7034179687500001, f(c)=9.40493604563164e-5, |b-a|=0.0001953
1250000000007
itr=12, a=-0.7035156250000001, b=-0.7034179687500001, c=-0.7034667968750001, f(c)=1.189811059454371e-6, |b-a|=9.
765625000000003e-5
Root found at c = -0.7034667968750001 with f(c) = 1.189811059454371e-6
(-0.7034667968750001, 1.189811059454371e-6)
```

Figure 4: Terminal output showing results of root finding for $e^x - x^2$ using the bisection method.

Problem Statement: Implement and analyze the Ridders' method for solving the equation $f(x) = 0$. For guidance, you might wish to read the Wikipedia page on the Ridders' method. Also feel free to consult other sources.

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Part 4

Numerically, compare the convergence rate of both algorithms. Discuss the results. What is the approximate order of convergence for the Ridders' algorithm?

Solution:
